Kinect Lite: Denoising Results

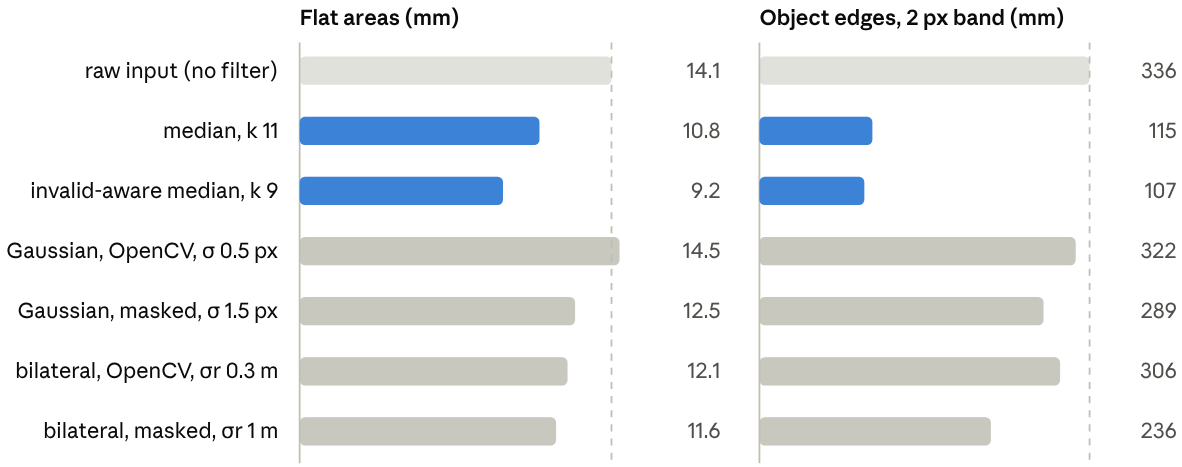
Oct 16, 2026 · @Sumanth G

Headline result

On 30 ICL test frames, the invalid-aware median cuts the average depth error from 26.2 mm to 12.8 mm, and both medians remove about two-thirds of the error at object edges. Gaussian and bilateral filters help flat areas less and edges much less.

Medians remove two-thirds of the edge error; averaging filters 4-30 %

Mean absolute error, 30 ICL test frames, one tuned setting per filter, 0.4 m gate. Blue = median filters.



Dashed lines = raw input. Bars start at 0 mm.

outputs/eval_selected/summary.csv · ICL frames 25:1509:50 (30 frames)

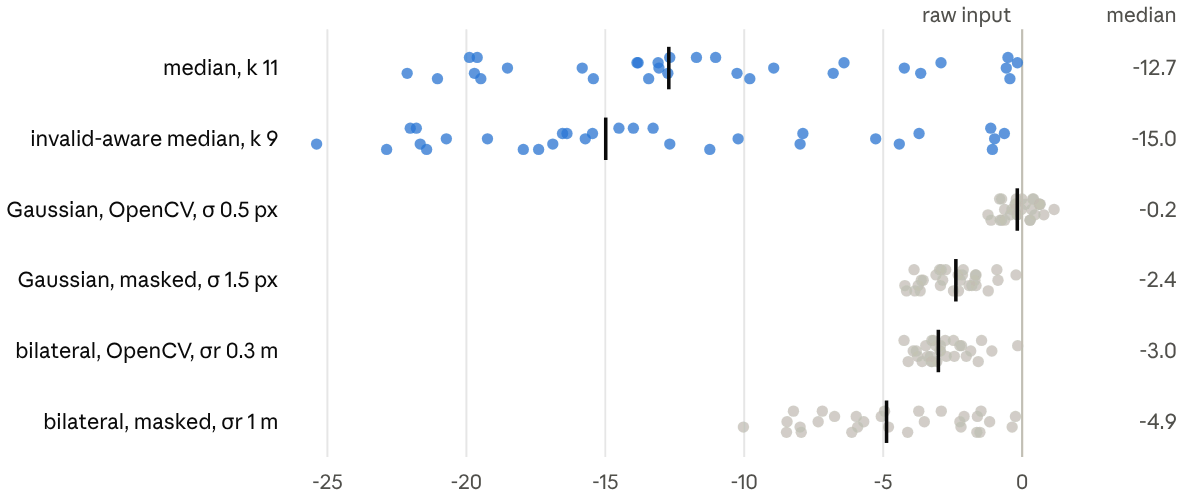
Error = mean absolute difference between filtered and true depth. Each filter uses the setting with the lowest error on separate tuning frames, with depth below 0.4 m treated as missing. The OpenCV Gaussian is the only one worse than raw on flat areas.

Frame by frame

The medians lower the error on every one of the 30 test frames, by up to 25 mm; the OpenCV Gaussian is worse than raw on 12 of them.

Medians lower the error on all 30 frames; OpenCV Gaussian barely changes it

Change in mean absolute error vs the raw input, one dot per ICL test frame (mm). Left of 0 = better than raw.



Black tick = median over frames. Frames 975-1125 show small changes for every filter.

outputs/eval_selected/per_frame.csv · 30 ICL test frames

How much a filter helps depends on the view: frames with many object edges in sight gain the most. Around frames 975-1125 the camera sees few edges and every filter changes the error by less than 2 mm.

How the settings were chosen

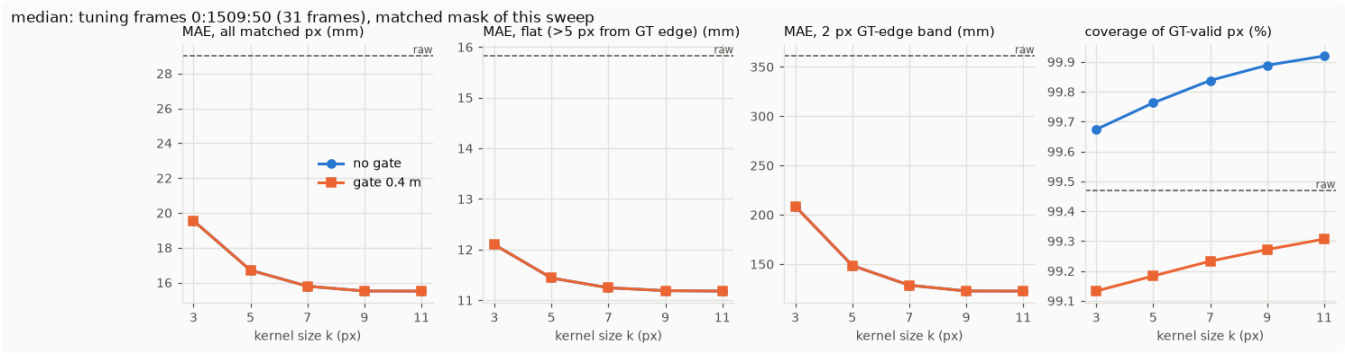
Each filter's setting was picked as the lowest overall error on 31 tuning frames (0, 50, 100 …), then scored once on the 30 test frames above. The tuning gains held on the test frames to within 6 percentage points.

In the median and Gaussian figures, orange = with the 0.4 m gate and blue = without; the dashed line is always the raw input.

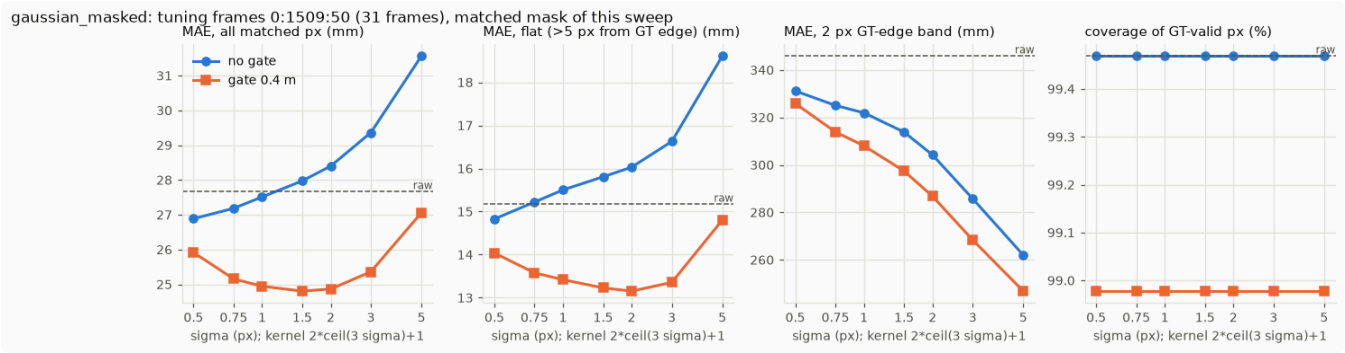
Invalid-aware median: error falls up to k = 9-11, then flattens.



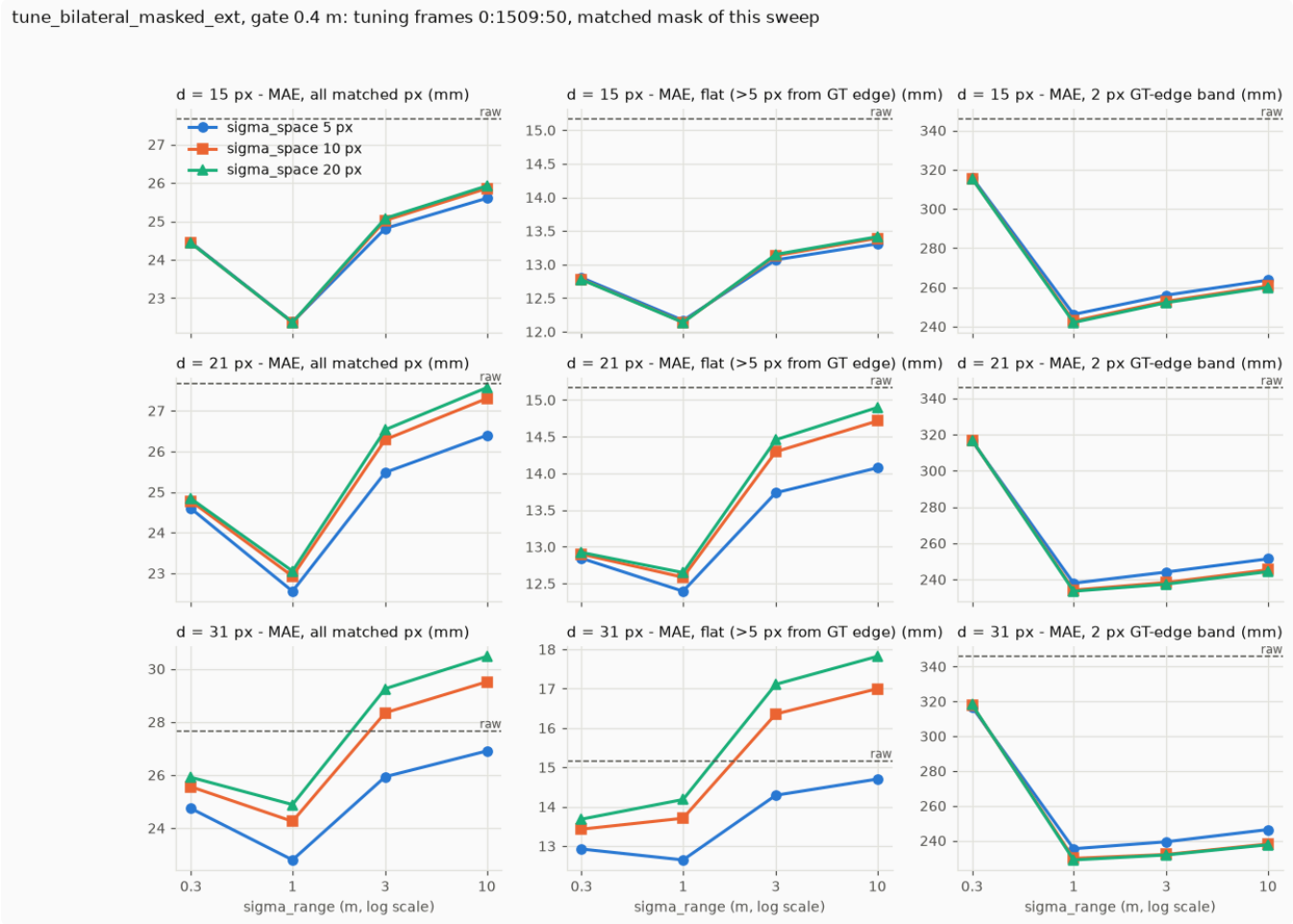
Plain median: with and without the gate are identical on the scored pixels. Where a gated value changes the output, the output is missing and drops out of scoring; the gate's effect shows only in coverage.



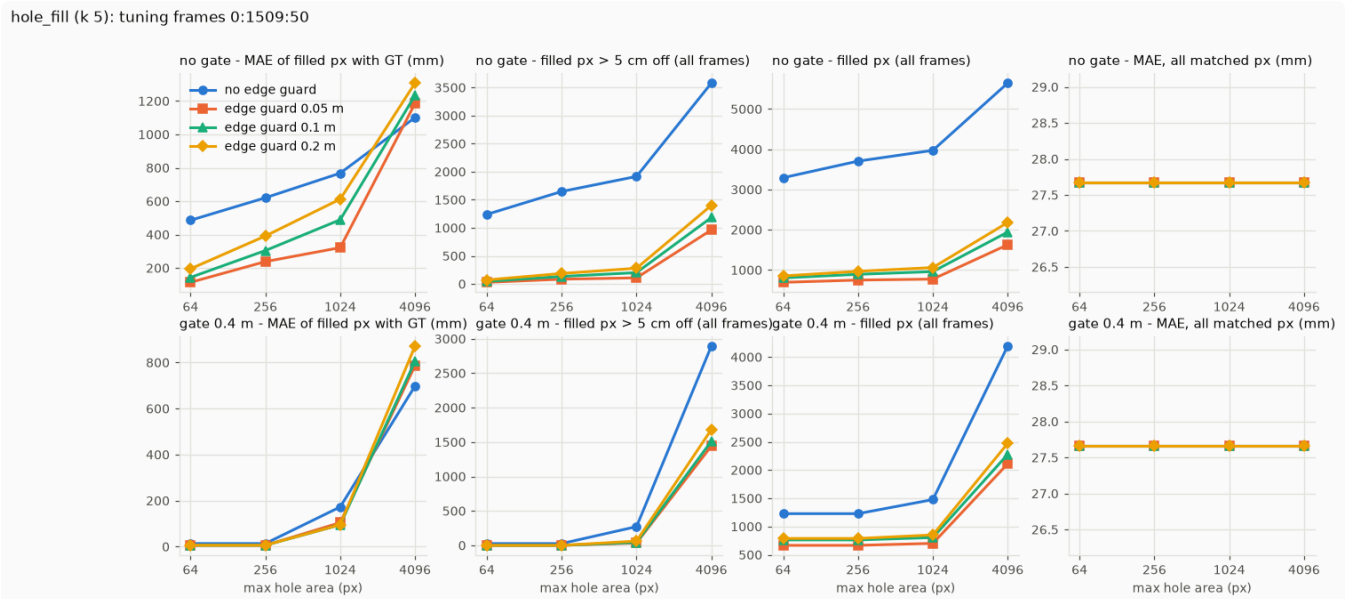
Masked Gaussian: without the gate it gets worse as sigma grows (it spreads the near-zero values); with the gate it is best at sigma 1.5-2 px.



Masked bilateral (with gate), extended grid: overall and flat error are lowest at a 15 px window; edge error keeps falling with larger windows, so this is a trade-off, not a single best.



Hole filling: it changes only empty pixels, so its error on scored pixels equals raw. Larger holes give more fills but more wrong fills; the edge guard cuts wrong fills by up to 40x.

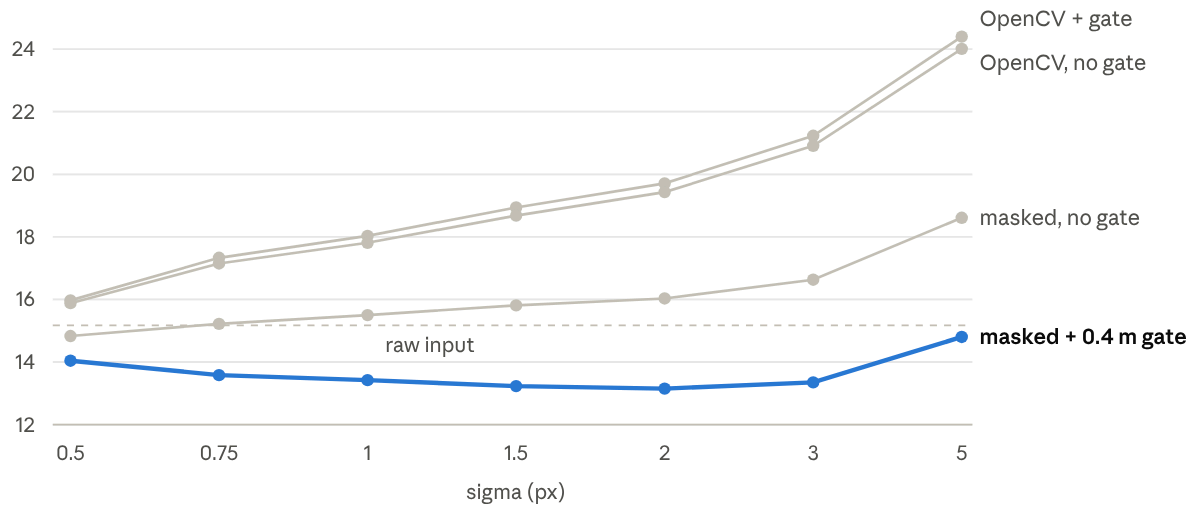


Missing pixels and the 0.4 m gate

How a filter treats missing pixels matters as much as which filter it is: the same Gaussian goes from worse than raw to better than raw on flat areas.

Same Gaussian: worse than raw when holes count as 0 m, better when left out

Error on flat areas (mm) vs blur width sigma, 31 ICL tuning frames. Lower is better.

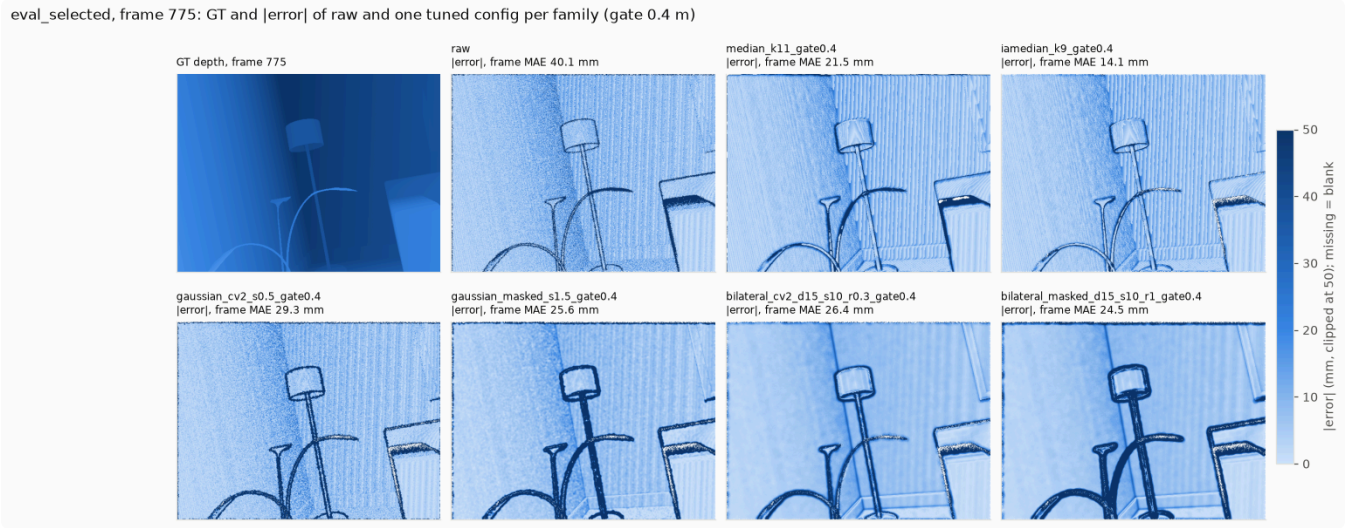


outputs/tune_gaussian_cv2 and tune_gaussian_masked summary.csv · 31 ICL tuning frames

- **OpenCV versions** need a number in every pixel, so holes enter as 0 m and drag nearby depth toward the camera. The gate makes this slightly worse, because it creates more holes.
- **Masked versions** leave holes out of the average. Without the gate they still spread the values below 0.4 m (which carry 25 % of the raw error); with the gate those are gone.
- **Medians** barely care: a 0 m or 0.1 m value is out-voted either way.

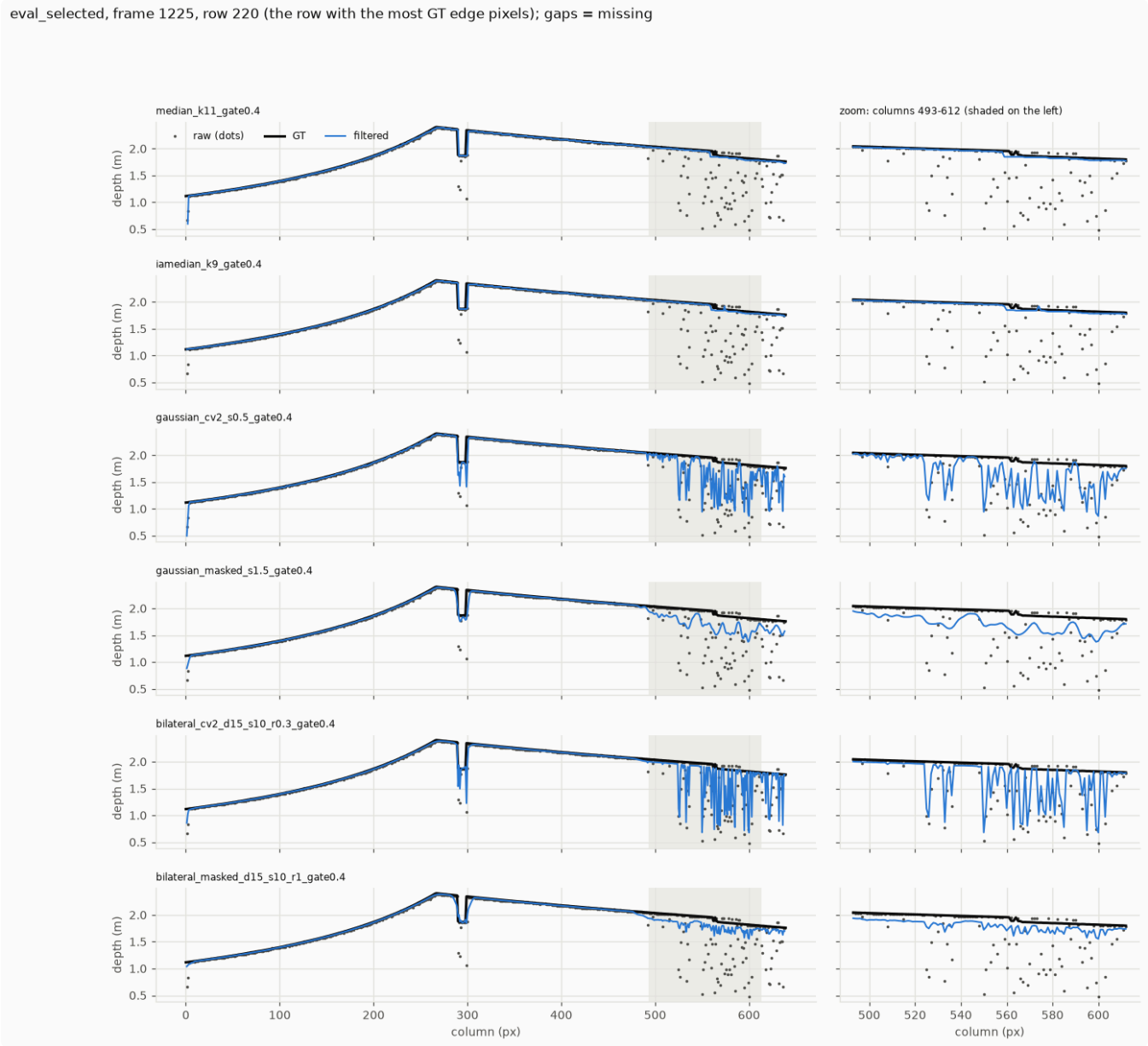
Seeing the errors

The medians' advantage is visible in the images: their error maps lose most of the dark outlines that the raw input and the averaging filters keep.



Each panel is |filtered depth - true depth| for ICL frame 775 (light = small, dark = 5 cm or more). The averaging filters thicken the dark outlines around the lamp and the plant; the medians thin them.

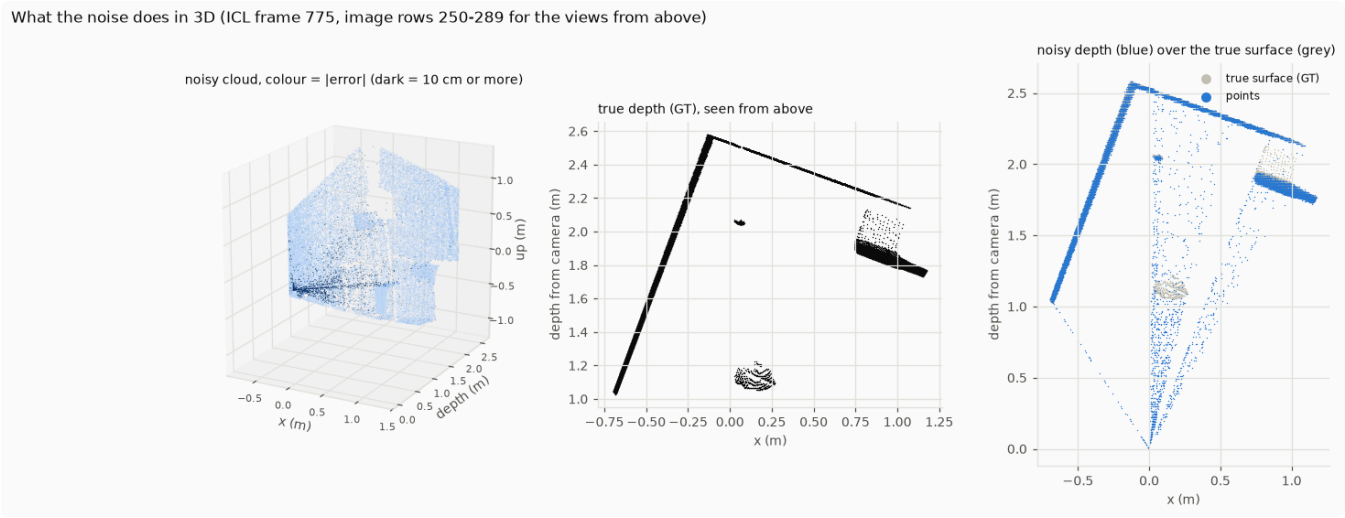
One image row, up close. The black line is the true depth along the row with the most edges in frame 1225; grey dots are the raw input; blue is each filter. The right-hand zoom is the shaded stretch.



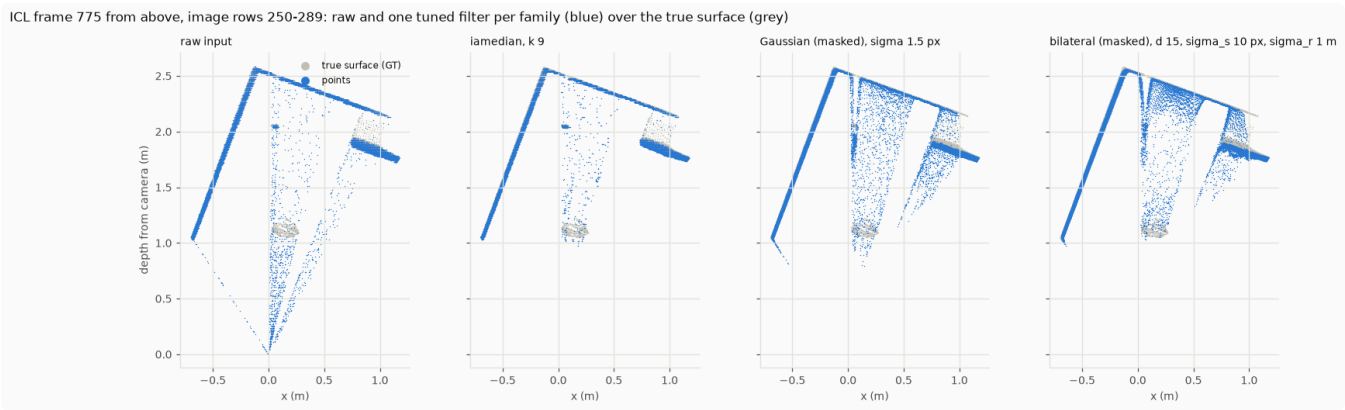
On the flat wall at columns 500-640, the raw input has scattered values 0.3-1.5 m too near. The medians follow the true depth; the Gaussian and bilateral are pulled 0.1-0.5 m toward those values.

In 3D

Seen as a point cloud from above, the noise is not random: the bad points line up on rays from the camera toward object edges, and the filters either remove those rays (median) or smear them into fans (Gaussian, bilateral).



Left: the noisy cloud, coloured by error. Middle and right: a 40-row band of the image seen from above, true surface vs noisy points. The camera sits at (0, 0).



The same band after each filter (blue) over the true surface (grey). The invalid-aware median leaves a few stray points; the Gaussian and bilateral pull the rays into fans behind the lamp and the sofa.

To rotate these clouds yourself, open `RESULTS/3d_viewer_icl.html` in a browser.

Real sensor: Asus Xtion (TUM)

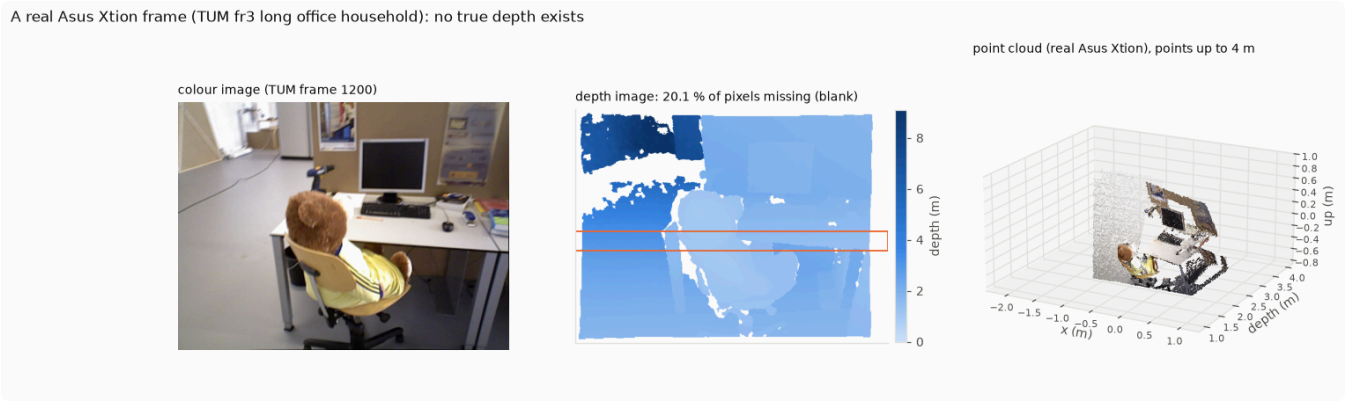
On real data from an Asus Xtion (a Kinect-type sensor), the bilateral and masked Gaussian make flat surfaces 8-11 times smoother, while the medians barely smooth them. With no true depth, we cannot say which edge changes are fixes and which are damage.

Real sensor (TUM): bilateral and masked Gaussian smooth flat areas 8-11x; medians 1.3x

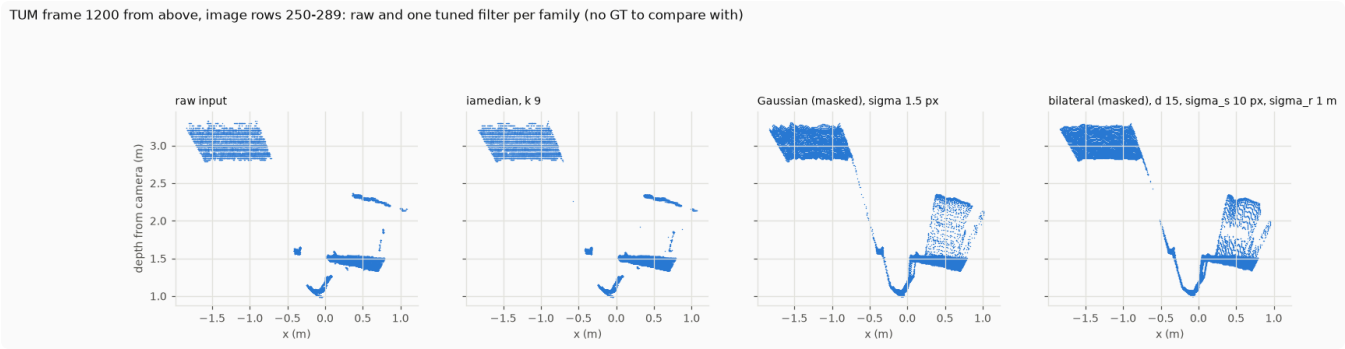
TUM fr3, 50 frames, no true depth: this shows what changes, not what is correct.



outputs/tum_selected/summary.csv · TUM fr3 frames 25:2509:50 (50 frames)



TUM frame 1200: 20 % of pixels have no depth (white), against about 0.5 % on ICL.



From above, the median keeps the gap between the desk (1.5 m) and the wall (2.3 m) empty. The Gaussian and bilateral fill it with new points: flying pixels the filter creates. The bilateral settings were tuned on ICL (σr = 1 m), which is too wide for real edges.

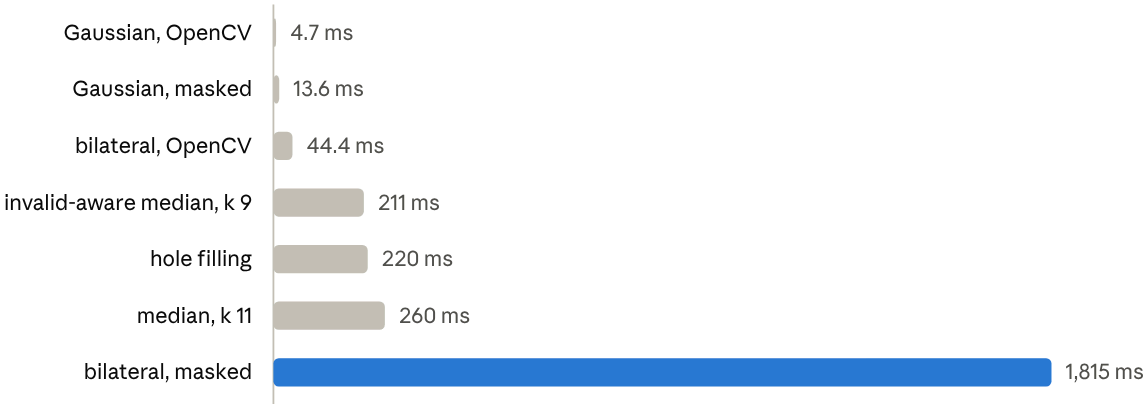
The 0.4 m gate changes nothing on TUM: no TUM frame has depth below 0.4 m.

Run time

All filters except the masked bilateral take under 0.3 s per frame; the OpenCV filters take 5-45 ms.

Masked bilateral takes 1.8 s per frame; every other filter under 0.3 s

Median time to filter one 640 x 480 frame on this PC (ms), Python. Bars start at 0.



The masked bilateral and the invalid-aware median are plain NumPy reference code, not optimised.

outputs/eval_selected/summary.csv (runtime_ms_median) · ICL, 30 frames

A Kinect delivers 30 frames per second (33 ms per frame), so only the OpenCV Gaussian and the masked Gaussian are fast enough for live use as written.

Limits of these results

The numbers are solid for ICL's simulated noise; how far they carry to a real camera is not measured.

- **Simulated noise.** ICL's edge spikes look like a side effect of how the noise was simulated. The medians' large edge advantage may be smaller on real data.
- **Alignment.** The noisy and true ICL images are about 1 px out of line, varying across the image. This adds about 1.5 mm to every flat-area score, and more at edges.
- **Same video.** Tuning and test frames are 25 frames apart in one sequence, so the test is fair but not independent.
- **Selection metric.** Settings were picked by overall error, which edge pixels dominate. Picking by flat-area error would choose different settings.

- **No true depth on TUM.** On real data we show what changes, not what is correct.

The theory behind all of this is in the companion document, Kinect Lite: Depth Denoising.